ENGINEERS WALLAH

CHEMISTRY

Lecture XI

Metals and Non-metals



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TOPICS *to be covered*

1. Physical Properties of Metals and Non-metals
2. Chemical Properties of Metals
3. Important Ores of Metals
4. Corrosion, Its Types and Ways to Prevent It

Knowledge Ride On



Physical Properties of Metals and Non-metals

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Chemical Properties of Metals

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Important Ores of Metals

Knowledge Ride On



Corrosion, Its Types and Ways to Prevent It

Knowledge Ride On



Efficiency Hacks

Glossary

Metal (धातु)
Non-metal (अधातु)
Malleability (आघातवर्धनीयता)
Ductility (तन्यता)
Hardness (कठोरता)
Sonority (ध्वन्यात्मकता)
Metallic lustre (धात्विक चमक)
Ore (अयस्क)
Corrosion (संक्षारण)
Alloy (मिश्रधातु)
Melting point (गलनांक)

Metallurgy (धातुकर्म)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which among the following element is the most abundant element in the universe?

SSC JE 2021

A Hydrogen , Helium

B Oxygen

C Nitrogen

D Helium

Concept Bites – Physical Properties of Metals and Non-metals

True or False

Are mercury and bromine the only metals found in liquid state at room temperature? NO!

Room Temp. (25 °C)

(METAL) Mercury → Liquid

(NON METAL) Bromine → Liquid

→ Only elements in liquid state at R.T.

Physical Properties of Metals and Non-metals

(धातु) (अधातु)

Property	Metals	Non-metals
Hardness (कठोरता)	Generally hard Exception: Lithium, Sodium and Potassium	Generally soft Exception: Diamond (form of carbon)
Lustre (चमक)	Lustrous	Generally non-lustrous ('dull') Exception: Iodine and graphite (form of C)
Malleability (आघातवर्धनीयता)	Generally, Malleable Exception: Mercury	Non-malleable

easily cut with a knife

resistance offered by a substance against: → CUTTING → SCRATCHING

∵ they reflect light

ability of metals (materials) to be converted into thin sheets on application of compressive force. (दबाव)

Physical Properties of Metals and Non-metals

ability to be converted into thin wires on application of tensile force. (खिंचाव)

(ध्वन्यात्मकता)

Property	Metals	Non-metals
Ductility (तन्यता)	Ductile Exception: Mercury	Generally non-ductile Exception: Carbon fibre
Sonority	Sonorous	Non-sonorous
Electrical conductivity	Good electrical conductors (because of free electrons)	Generally poor electrical conductors Exception: Graphite

Physical Properties of Metals and Non-metals

Property	Metals	Non-metals
Thermal conductivity	Generally good thermal conductors Exception: Lead (Pb) & Mercury (Hg)	Generally poor thermal conductors Exception: Diamond
Melting point	High melting point Exception: Gallium and Caesium	Generally, low melting point Exception: Diamond

→ Role of e^- (negligible)

→ Atomic vibrations (PHONONIC HEAT TRANSFER)

Gallium: m.p. 29.7 °C

Caesium: 28.4 °C

Melts on human palm (37 °C)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Malleability is the property of metals _ _ _ _ _ _ _ _.

RRB JE 2015

A to be drawn in wires (DUCTILITY)

B to be converted into sheets

C to conduct electricity

D To conduct heat

Question

Which of the following element is lustrous in nature?

RRB JE 2015

A Carbon (अधात्)

B Iodine (धात्)

C Sulphur (-11-)

D Nitrogen (-11-)

Question

Give an example of a metal which is a liquid at room temperature.

RRB JE 2019

A Sodium

B Iodine

C Mercury

D Calcium

Question

Give an example of a metal which can easily be cut with a knife.

RRB JE 2019

A Silver

B Mercury

C Iron

D Sodium

Concept Bites – Chemical Properties of Metals and Non-metals

Reaction of Metals with Oxygen

Metal + Oxygen ⟶ Metal oxide

R.T. (25°C) $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$

WHITE LAYER

Strongly Heating

$2Mg(s) + O_2(g) \longrightarrow 2MgO(s) + heat + light$

(WHITE POWDER)

↓

White dazzling flame

Reaction of Metals with Water

C-I

Metal + Water ⟶ Metal oxide + Hydrogen

C-IV

$Na + H_2O \longrightarrow Na_2O + H_2$ (catches fire) + heat

[Reactivity of Na > H]

C-II Dissolves in Water

Na_2O → NaOH

Metal hydroxide

C-III

- $\rightarrow Na_2O$
- $\rightarrow K_2O$
- $\rightarrow CaO$
- $\rightarrow MgO$

→ Soluble in H_2O → Metal hydroxide

Reaction of Metals with Acids

(aq): Dissolved in water
(↑): Evolved gas

Metal + Dil. Acid ⟶ Salt + Hydrogen

⇒ $Zn(s) + 2\,dil.HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(\uparrow)$

⇒ $Fe(s) + 2\,dil.HCl(aq) \rightarrow FeCl_2(aq) + H_2(\uparrow)$

$Ag + dil.\ HCl \rightarrow$ NO Rx^n

K	Kudi	Potassium
Na	Naal	Sodium
Ca	Cor	Calcium
Mg	Maango	Magnesium
Al	Alto	Aluminium
Zn	Zisko	Zinc
Fe	Fir	Iron
Pb	Lekar	Lead
H	Hum	**Hydrogen**
Cu	Chale	Copper
Hg	Mathura	Mercury
Ag	Satya	Silver
Au	Ghumne	Gold

Reaction of Metals with Acids

Mn: Manganese

Metal (Mg and Mn) + dilute nitric acid -> Salt + Hydrogen gas

diluted

$$Mg + 2HNO_3 \longrightarrow Mg(NO_3)_2 + H_2$$

$$Mn + 2HNO_3 \longrightarrow Mn(NO_3)_2 + H_2$$

Other Metals + dilute nitric acid -> Salt + Water + NO / N_2O / NO_2

reduces to

Strong oxidising agent/oxidant

oxidises H_2 to

Thermit Reaction

Reactivity of Al > Fe

Displacement + Exothermic

$$Fe_2O_3(s) + 2Al(s) \xrightarrow[\text{Mg ribbon}]{\text{Ignited with}} \underset{\text{Molten}}{2Fe(l)} + Al_2O_3(s) + \text{Heat}$$

because of this

The reaction is therefore used for welding the broken parts of *iron machinery, railway tracks* etc. The reaction is known as thermite reaction.

Aqua Regia

Aqua regia, which in Latin means 'royal water' is a mixture of concentrated nitric acid and hydrochloric acid in the ratio of 1:3 by volume. It is highly corrosive and fuming liquid.
It is one of the few reagents that can dissolve gold and platinum.

free state / native state in earth's crust

Reaction of Non-metals with Oxygen and Water

C-I

$C + O_2 \rightarrow CO_2$

$S + O_2 \rightarrow SO_2$

Non-metal + Oxygen → Non-metal oxide

C-II

Non-metal + Water → NO RXN

Non-metal can't displace hydrogen (non-metal) in water

C-III

$CO_2 + H_2O \longrightarrow H_2CO_3$ (Carbonic acid)

Non-metal Oxide + Water → Acid

C-IV

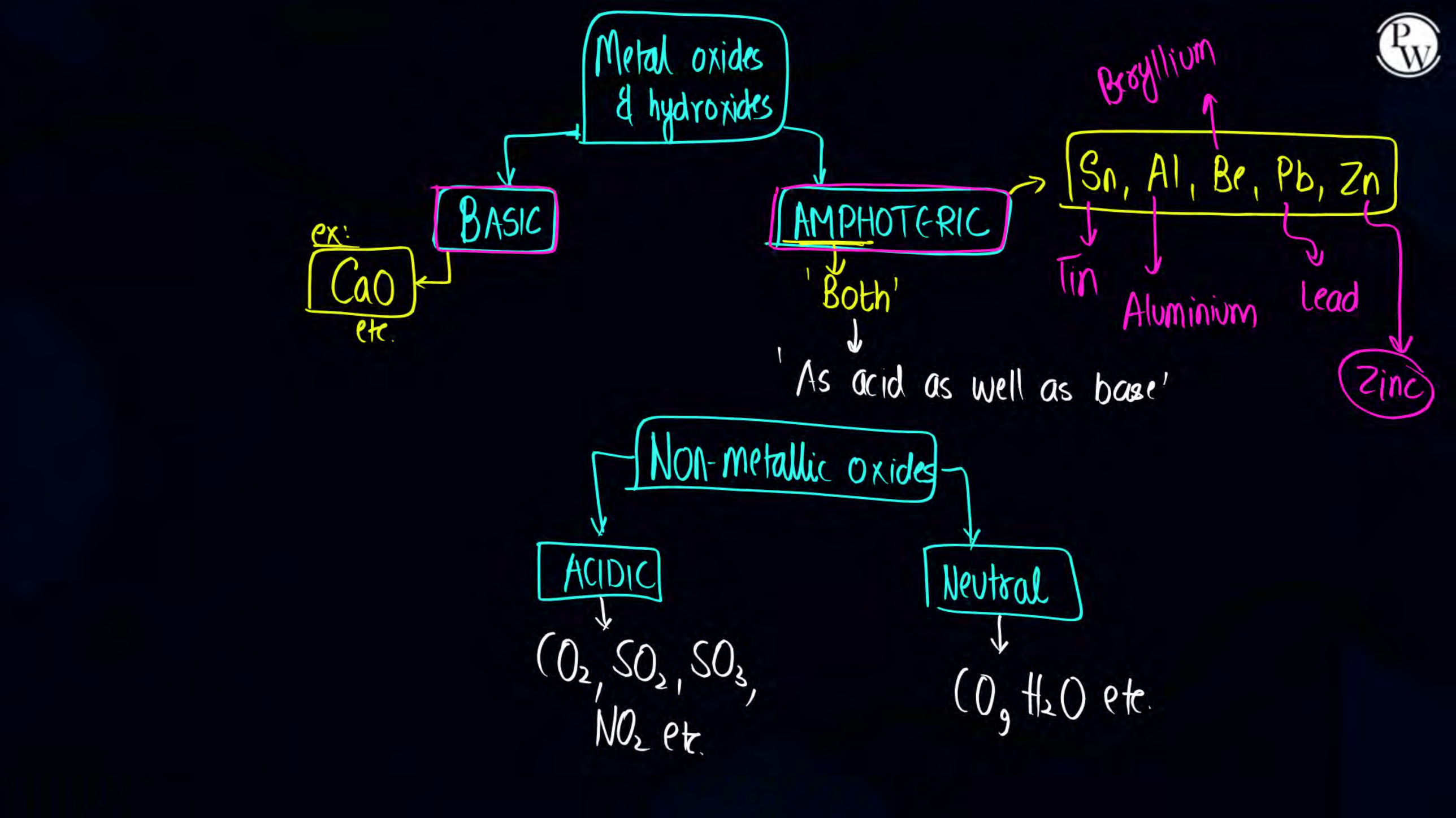
ACID ANHYDRIDE → CO_2

double acid anhydride

$SO_2 + H_2O \longrightarrow H_2SO_3$ (Sulphurous acid) → SO_2

$SO_3 + H_2O \longrightarrow H_2SO_4$ (Sulphuric acid) → SO_3

$NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$ (nitrous acid, nitric acid) → NO_2


Metal oxides & hydroxides
BASIC
ex:
CaO
etc.
AMPHOTERIC
'Both'
'As acid as well as base'
Sn, Al, Be, Pb, Zn
Tin
Aluminium
Beryllium
Lead
Zinc
Non-metallic oxides
ACIDIC
CO_2, SO_2, SO_3, NO_2 etc.
Neutral
CO, H_2O etc.



Yes or No

Do non-metals react with acids?

Non-metal + Acid ⟶ No rxn

<Non-metal can't displace hydrogen of acids>

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which among the following is an amphoteric oxide?

RRB JE 2015

A Na_2O

B SO_3

C Ag_2O

D ZnO

Question

A metal forms an amphoteric oxide on reaction with oxygen. The metal is:

A Al

B Na

C K

D Mg

Question

An element X contains only one electron in its outermost shell. Which of the following statement(s) are correct about X?

(i) It forms an acidic oxide ✗

(ii) It forms a basic oxide. ✓

(iii) It is a good conductor of electricity. ✓

(iv) It has a dull and non-lustrous surface. ✗

lustrous surface

RRB JE 2015

K L M

Na 2 8 1

Na_2O

BASE

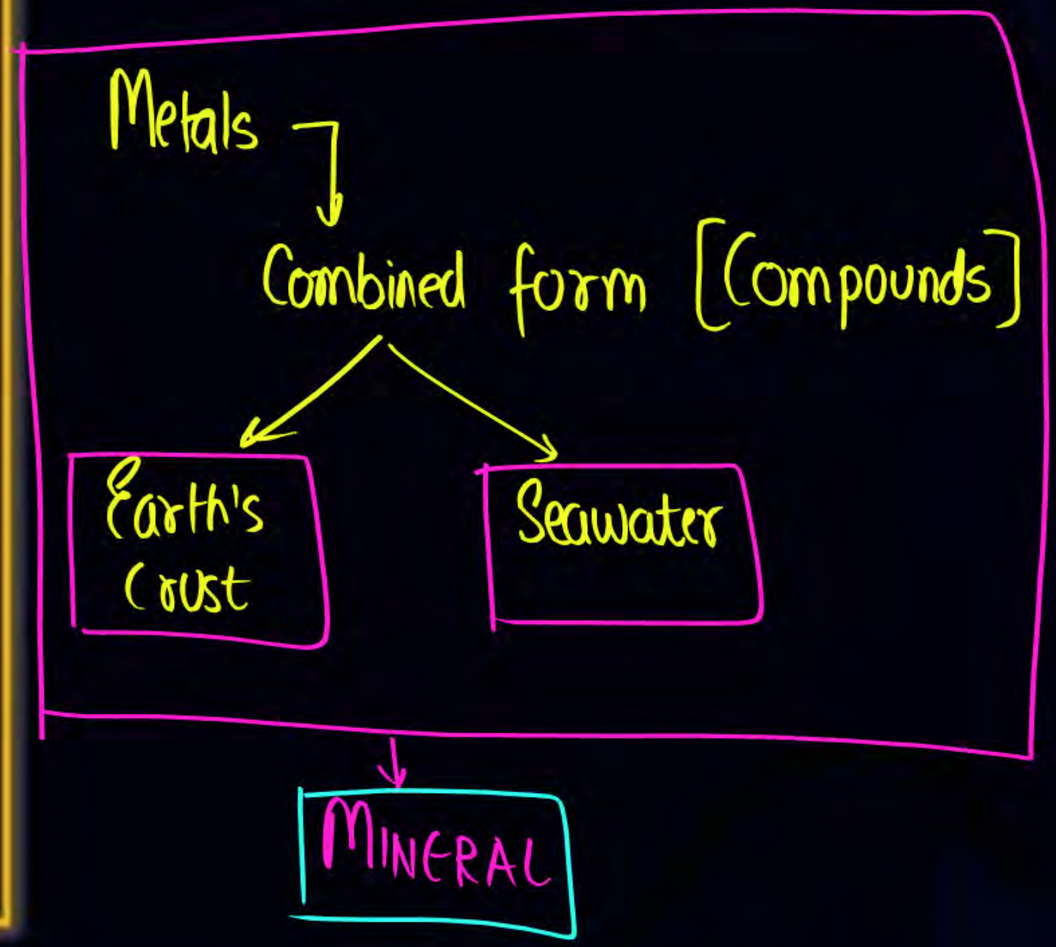
A. (i) and (iiii)

B. (ii) and (iii)

C. (i) and (iv)

D. (iii) and (iv)

Concept Bites – Important Ores of Metals



	Metal present	Mineral	Ore
Clay	Al	✓	✗
Bauxite	Al	✓	✓

∵ metal can be extracted from bauxite economically and conveniently.

Earthly Impurities (sand, limestone etc.) stuck on ore are called gangue or matrix

Ores of Common Metals

Metal	Name of the ore	Chemical formula of ore
(a) Sodium (Na) →	Rock salt	NaCl
(b) Aluminium (Al) →	Bauxite	$Al_2O_3.2H_2O$
(c) Zinc (Zn) →	(i) Zinc blende (ii) Calamine	(i) ZnS (ii) $ZnCO_3$
(d) Iron (Fe) →	(i) Haemetite (ii) Magnetite	(i) Fe_2O_3 (ii) Fe_3O_4 ($FeO.Fe_2O_3$)
(e) Copper (Cu) →	Copper glance	Cu_2S (ii) Cuprite (Cu_2O)
(f) Mercury (Hg) →	Cinnabar →	HgS

(iii) Iron pyrite (FeS_2)

(iii) Copper pyrite ($CuFeS_2$)

Practice and PYQ's Wallah



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Question

Fe_2O_3 is the chemical formula of an iron ore called:

SSC JE 2019

A Limonite

B Iron pyrite

C Magnetite → (Fe_3O_4 / $FeO.Fe_2O_3$)

D Haematite

Question

Galena is basically a _____.

SSC JE 2021

Galena → ore of lead
(PbS)

A chloride

B fluoride

C hydride

D sulphide

Question

Galena is an ore of:

RRB JE 2015

A Lead (Pb)

B Copper

C Aluminium

D Iron

Question

Which of the following is the primary constituent of naturally occurring mineral magnetite on Earth?

Fe_3O_4

SSC JE 2022

A Calcium oxide

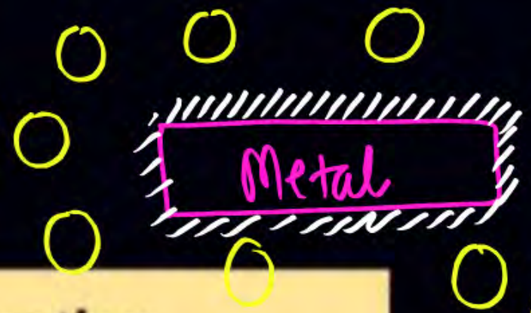
B Iron oxide

C Aluminium oxide

D Magnesium oxide

Concept Bites – Corrosion, Its Types and Ways to Prevent It

Corrosion (संक्षारण) and Its Types



Atmospheric gases (non-metallic)

Type of corrosion	Colour of layer and chemical equation
Rusting of iron Iron reacts with moist air to form hydrated ferric oxide.	 Reddish-brown ⇝ RUST $4Fe(s) + 3O_2(g) + xH_2O(g) \rightarrow 2Fe_2O_3.xH_2O(s)$

Corrosion
- Rusting → Iron (Fe) → RUST
- Tarnishing → Cu, Zn, Mg, Pb, Ag etc. → PATINA

RUST ← flaky (non-sticky) ← पपड़ी बनकर गिरेगी

PATINA → Protective layer as it is non-flaky

Corrosion (संक्षारण) and Its Types

Type of corrosion	Colour of layer and chemical equation
Tarnishing of copper Copper reacts with moist (H_2O) carbon dioxide (CO_2) gas in the air to (O_2) form copper carbonate.	 Statue of liberty Green PATINA Copper carbonate hydroxide $2Cu(s) + CO_2(g) + O_2(g) + H_2O(g) \rightarrow CuCO_3.Cu(OH)_2(s)$

Corrosion (संक्षारण) and Its Types

Type of corrosion	Colour of layer and chemical equation
Tarnishing of silver Silver reacts with hydrogen sulphide gas present in the air to form silver sulphide.	 Black PATINA $4Ag(s) + 2H_2S(g) + O_2(g) \rightarrow 2Ag_2S(s) + 2H_2O(g)$

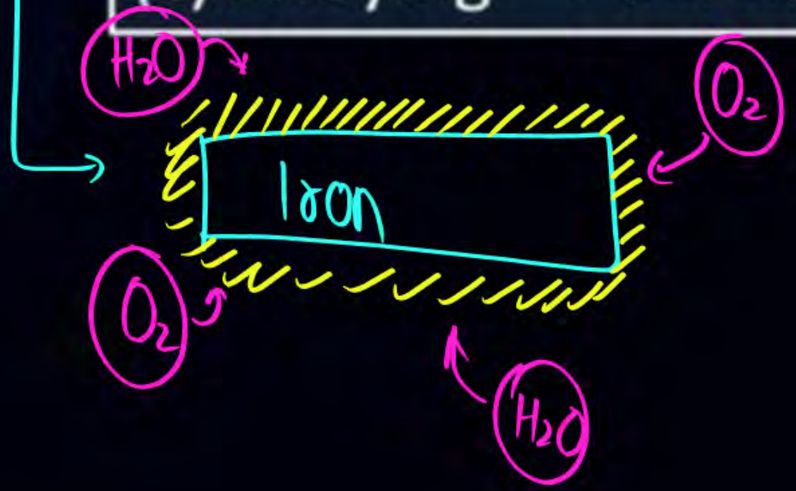
Ways to Prevent Rusting

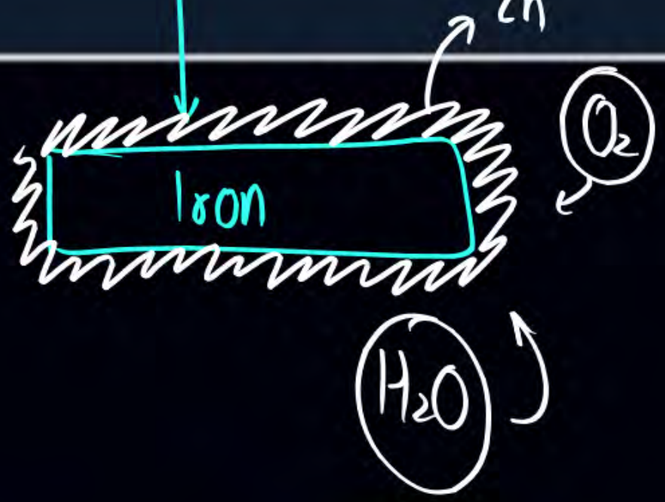
The below methods are used to prevent the rusting of iron:

(a) Barrier Protection *(Applying paint, oil or grease)*

(b) Sacrificial Protection *(Galvanisation – Applying a layer of Zinc on Iron)*

(c) Alloying → 'Permanent Solution'





(i) Barrier

(ii) Reacts with O_2 & H_2O on behalf of iron ∵ it is more reactive.

Ways to Prevent Rusting

Alloy → (मिश्रधातु)

Pure iron is soft, stretches easily when hot and prone to rust.
Iron is alloyed with carbon and it becomes hard and strong.

↳ 'Steel'

More to Know:

Stainless Steel: Iron + Chromium + Nickel

- Nickel → -Provides hardness.
- Iron → -Not prone to rusting. -Used to create utensils.
- Chromium → -Resistance to corrosion. -Prevents from getting stains.

More About Alloys (मिश्रधातु)

Alloy	Composition	Properties	Important Uses
(i) Brass (पीतल)	Copper (80%) and Zinc (20%)	Malleable, strong, resists corrosion	Utensils, screws etc.
(ii) Bronze (कांस्य)	Copper (90%) and Tin (10%)	Strong and resists corrosion	Coins, bells etc.
(iii) German silver (No silver)	Copper (60%), Zinc (20%) and Nickel (20%)	Shines like silver, resistant to corrosion	Electroplating
(iv) Solder	Lead (50%) and Tin (50%)	Lower melting point than lead and tin	For joining electrical wires together.

More About Alloys (मिश्रधातु)

ex: Dental amalgam → Ag + Hg (Silver, Mercury)

Note:

- The alloys of mercury with any other metal is called amalgam.
- Pure gold is 24 carat which is soft and hence, it is not used to make jewellery. In India, 22 carat gold is ornamental gold. This means 22 parts by weight of gold is alloyed with 2 parts by weight of either copper or silver.

↓ for Strength

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which among the following alloy contains tin?

RRB JE 2014

A Brass

B Solder → (Lead & Tin)

C Duralumin

D Steel

Question

Which is the metal mainly used for galvanizing iron buckets for our daily life?

RRB JE 2015

A Mercury

B Cadmium

C Tin

D Zinc

Efficiency Hacks!

Ways to Increase Memory or Recall Power

- **3:** Don't eat anything 3 hours before sleep
- **2:** Meditation/positive thoughts 2 hours before sleep.
- **1:** Keep your gadgets away 1 hour before your sleep.




#sbsathhai
#pwsathhai
Thank
You !